

Storage Units -- Guided Notes

Unit tp-1.5 | CompTIA Network+ N10-009 Obj. FC0-U71 1.3 | N10-008 FC0-U71 1.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The smallest unit of digital data is the _____, which holds a single binary value (0 or 1). Eight bits make one _____. A _____ stores 1,024 bytes, and a _____ stores 1,024 kilobytes.

Question 2

Fill in the storage unit table:

1 KB = _____ bytes

1 MB = _____ KB

1 GB = _____ MB

1 TB = _____ GB

Question 3

Hard drive manufacturers advertise storage using the _____ (decimal/binary) definition where 1 GB = 1,000,000,000 bytes. Operating systems measure storage using the _____ (decimal/binary) definition where 1 GB = 1,073,741,824 bytes. This is why a 500 GB drive appears as approximately _____ GB in Windows File Explorer.

Question 4

A short text message is approximately 160 _____ (bits/bytes). A typical MP3 song file is around 4 to 5 _____. A 4K movie might be 50 to 100 _____. Choose the correct unit for each: bits, bytes, megabytes, gigabytes.

Question 5

_____ is the space used to run programs and store data while the computer is on (RAM). _____ is the space used to permanently save files, programs, and the OS (hard drive/SSD). The key difference is that memory is _____ (lost when power is off) while storage is _____.

Question 6

A file is 750 MB. A flash drive has 1 GB of free space. Will the file fit? Show your reasoning:

1 GB = _____ MB

750 MB _____ (is/is not) less than 1,024 MB

Conclusion: the file _____ fit on the drive.

Question 7

When choosing appropriate storage units, match each description to the correct unit:

A single character of text: _____

A Word document (5 pages): _____

A smartphone photo: _____

A Blu-ray movie: _____

Question 8 -- Real World Application

REAL WORLD: A student buys an external hard drive advertised as 2 TB. When they plug it in, Windows shows only 1.81 TB of usable space. The student believes they were sold a defective or undersized drive and wants to return it. Explain what is actually happening, using the decimal vs. binary definition of gigabyte, and whether the drive is defective.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-1-5-storage-units/tp-1-5-storage-units-answer-key.pdf
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